Build pipeline model 🛠️

The objective of this notebook was to build a regression model to predict the price of diamonds based on their physical and visual characteristics. To achieve this goal, we will create and train pipeline as well as testing it. The dataset was loaded, preprocessed, and then we applied pipeline models to predict the price. To ensure the best model is selected, we applied GridSearchCV and observed the error metrics and R2 values.

Importing Libraries 🛠️

We will start by importing the necessary libraries for building a pipeline model. We will be using the following libraries:

- `pandas` - for data manipulation and analysis
- `numpy` - for numerical operations
- `dill` - for saving the model
- `train_test_split` - for splitting the dataset into training and testing sets
- `Pipeline` - for building a pipeline of transformers and estimators
- `SimpleImputer`, KNNImputer - for imputing missing values
- `ColumnTransformer` - for column-wise transformation
- `OneHotEncoder` - for categorical encoding
- `BaseEstimator`, TransformerMixin - for creating custom transformers
- `CatBoostRegressor` - for building a CatBoost regression model
- `Pool` - for creating a CatBoost data pool
- `GridSearchCV` - for hyperparameter tuning

In [ ]:
```
import pandas as pd
import numpy as np
import dill
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer, KNNImputer
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.base import BaseEstimator, TransformerMixin
from catboost import CatBoostRegressor, Pool
from sklearn.model_selection import GridSearchCV
```

Loading Data 📁

Next, we will load the dataset using pandas read_parquet function.

In [ ]:
```
df = pd.read_csv("./../data/filtered_diamonds.csv")
```

Split the dataset into training and testing sets 📊

We will be using the `train_test_split` function from the `sklearn.model_selection` library to split the dataset into a training set and a testing set.

In [ ]:
```
X = df.drop("Price", axis=1)
y = df["Price"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Make pipeline 👷

This is a pipeline that consists of several transformers and an estimator. It is designed to preprocess and model diamond price data. The pipeline has four main stages, which are as follows:

- `imputer`: This transformer replaces missing values in the dataset. It contains two imputers - one for categorical features (using the most frequent value) and another for numeric features (using the KNN algorithm).
- `encoder`: This transformer encodes the categorical features using OneHotEncoder for the Shape feature and an OrdinalConverter for the remaining categorical features.
- `selector`: This transformer is a custom transformer that uses a CatBoostRegressor model to identify the most important features in the dataset. The transformer returns the top 14 features based on their importance.
- `regressor`: This is the final stage of the pipeline and it contains the CatBoostRegressor model. The model is used to predict the diamond prices using the selected features.

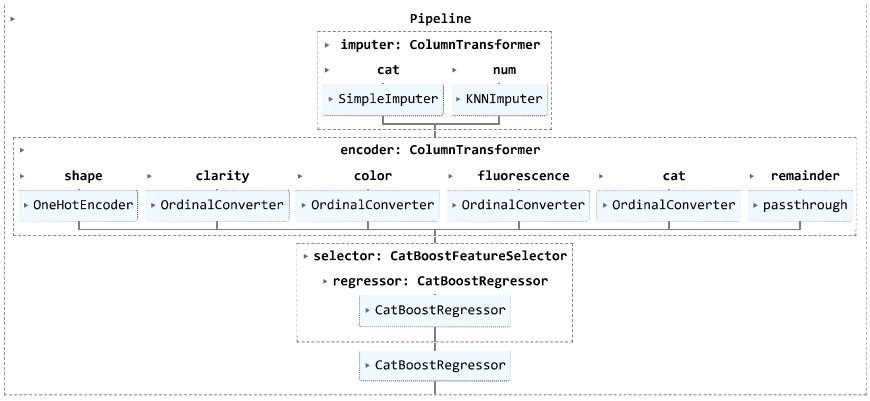
In [ ]:
```
class OrdinalConverter(BaseEstimator, TransformerMixin):
    def __init__(self, order):
        self.order = order
        self.map_func = np.vectorize(lambda x: self.order.index(x))

    def fit(self, X, y=None):
        return self

    def transform(self, X):
        return self.map_func(X)


class CatBoostFeatureSelector(BaseEstimator, TransformerMixin):
    def __init__(self, regressor, num_features):
        self.regressor = regressor
        self.num_features = num_features
        self.feature_importances_ = None

    def fit(self, X, y):
        pool = Pool(X, y, cat_features=[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13])
        self.regressor.fit(pool)
        self.feature_importances_ = self.regressor.get_feature_importance(
            pool, type="PredictionValuesChange"
        )
        return self

    def transform(self, X):
        sorted_features = sorted(
            range(len(self.feature_importances_)),
            key=lambda k: -self.feature_importances_[k],
        )
        top_features = sorted_features[: self.num_features]
        return X[:, top_features]

    def get_feature_importances(self, feature_names=None):
        if feature_names is None:
            feature_names = [
                f"feature_{i}" for i in range(len(self.feature_importances_))
            ]
        sorted_importances = sorted(
            zip(feature_names, self.feature_importances_), key=lambda x: -x[1]
        )
        return sorted_importances


imputer = ColumnTransformer(
    transformers=[
```

```
        (
            "cat",
            SimpleImputer(strategy="most_frequent"),
            ["Shape", "Clarity", "Colour", "Fluorescence", "Cut", "Polish", "Symmetry"],
        ),
        ("num", KNNImputer(n_neighbors=5), ["Weight", "Length", "Width", "Depth"]),
    ]
)

encoder = ColumnTransformer(
    transformers=[
        ("shape", OneHotEncoder(drop="if_binary", dtype=int), [0]),
        (
            "clarity",
            OrdinalConverter(
                ["I3", "I2", "I1", "SI2", "SI1", "VS2", "VS1", "VVS2", "VVS1", "IF"]
            ),
            [1],
        ),
        (
            "color",
            OrdinalConverter(
                [
                    "Y-Z",
                    "W-X",
                    "W",
                    "U-V",
                    "S-T",
                    "Q-R",
                    "O-P",
                    "O",
                    "N",
                    "M",
                    "L",
                    "K",
                    "J",
                    "I",
                    "H",
                    "G",
                    "F",
                    "E",
                    "D",
                    "FANCY",
                ]
            ),
            [2],
        ),
        (
            "fluorescence",
            OrdinalConverter(["N", "M", "F", "SL", "ST", "VS", "VSL"]),
            [3],
        ),
        ("cat", OrdinalConverter(["FR", "GD", "VG", "EX"]), [4, 5, 6]),
    ],
    remainder="passthrough",
)

selector = CatBoostFeatureSelector(
    regressor=CatBoostRegressor(verbose=0, allow_writing_files=False), num_features=14
)

regressor = CatBoostRegressor(verbose=0, allow_writing_files=False)


pipeline = Pipeline(
    steps=[
        ("imputer", imputer),
        ("encoder", encoder),
        ("selector", selector),
        ("regressor", regressor),
    ]
)
```

Train the model

In [ ]:
```
pipeline.fit(X_train, y_train)
```

Out[ ]:



Testing the model using train dataset

In [ ]:
```
pipeline.score(X_train, y_train)
```

Out[ ]:
```
0.9905597836052875
```

Testing the model using test dataset

In [ ]:
```
pipeline.score(X_test, y_test)
```

Out[ ]:
```
0.8827497503683435
```

Hyperparameters tuning

We will use GridSearchCV to hyperparameters tuning. Parameters which are tuning are `selector__num_features`, `regressor__learning_rate`, `regressor__depth` and `regressor__l2_leaf_reg`. with cv=3. And we will use r2 for scoring our performance

In [ ]:
```
# Define the hyperparameters to tune
params = {
    "selector__num_features": [10, 12, 14, 16, 18],
    "regressor__learning_rate": [0.01, 0.05, 0.1],
    "regressor__depth": [4, 6, 8],
    "regressor__l2_leaf_reg": [1, 3, 5],
}
```

In [ ]:
```
# Create the GridSearchCV object
grid_search = GridSearchCV(
    pipeline, param_grid=params, cv=3, scoring="r2", n_jobs=-1, verbose=3
)
```

In [ ]:
```
# Fit the GridSearchCV object to the data
grid_search.fit(X_train, y_train)
```

```
Fitting 3 folds for each of 135 candidates, totalling 405 fits
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=10;, score=0.868 total time=  30.3s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=10;, score=0.900 total time=  32.0s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=14;, score=0.861 total time=  31.6s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=12;, score=0.899 total time=  31.8s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=12;, score=0.868 total time=  32.3s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=10;, score=0.861 total time=  32.6s
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[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=14;, score=0.868 total time=  32.5s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=14;, score=0.898 total time=  31.1s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=18;, score=0.901 total time=  29.2s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=16;, score=0.860 total time=  30.3s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=18;, score=0.861 total time=  30.3s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=16;, score=0.868 total time=  30.7s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=10;, score=0.883 total time=  30.4s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=18;, score=0.867 total time=  31.3s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=16;, score=0.902 total time=  31.7s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=10;, score=0.917 total time=  29.7s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=10;, score=0.862 total time=  32.2s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=12;, score=0.910 total time=  30.7s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=14;, score=0.868 total time=  30.0s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=12;, score=0.876 total time=  31.6s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=12;, score=0.865 total time=  31.6s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=14;, score=0.910 total time=  30.2s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=14;, score=0.881 total time=  31.2s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=18;, score=0.881 total time=  29.1s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=16;, score=0.879 total time=  31.6s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=16;, score=0.909 total time=  30.2s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=16;, score=0.876 total time=  31.0s
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[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=10;, score=0.889 total time=  30.4s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=10;, score=0.866 total time=  30.8s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=18;, score=0.875 total time=  32.1s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=10;, score=0.903 total time=  31.1s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=16;, score=0.877 total time=  28.2s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=14;, score=0.914 total time=  29.0s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=12;, score=0.879 total time=  31.1s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=14;, score=0.881 total time=  30.4s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=12;, score=0.870 total time=  31.2s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=12;, score=0.910 total time=  31.0s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=14;, score=0.871 total time=  30.5s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=16;, score=0.878 total time=  29.0s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=10;, score=0.870 total time=  28.6s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=10;, score=0.857 total time=  29.1s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=16;, score=0.904 total time=  30.0s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=18;, score=0.914 total time=  30.5s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=18;, score=0.878 total time=  31.8s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=10;, score=0.896 total time=  30.8s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=18;, score=0.878 total time=  31.7s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=14;, score=0.857 total time=  28.7s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=12;, score=0.867 total time=  31.3s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=14;, score=0.869 total time=  29.9s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=12;, score=0.857 total time=  31.1s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=12;, score=0.896 total time=  30.8s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=14;, score=0.895 total time=  29.4s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=16;, score=0.856 total time=  31.1s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=16;, score=0.866 total time=  31.5s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=16;, score=0.897 total time=  28.4s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=18;, score=0.854 total time=  28.2s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=18;, score=0.898 total time=  28.8s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=10;, score=0.871 total time=  28.5s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=18;, score=0.866 total time=  30.4s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=10;, score=0.873 total time=  31.1s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=12;, score=0.881 total time=  29.7s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=10;, score=0.917 total time=  31.4s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=14;, score=0.883 total time=  27.8s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=12;, score=0.870 total time=  30.9s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=14;, score=0.916 total time=  29.1s
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[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=14;, score=0.864 total time=  31.1s
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[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=16;, score=0.906 total time=  29.5s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=16;, score=0.872 total time=  31.0s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=18;, score=0.880 total time=  28.8s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=10;, score=0.879 total time=  28.8s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=18;, score=0.909 total time=  29.1s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=10;, score=0.872 total time=  29.4s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=18;, score=0.875 total time=  31.5s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=10;, score=0.912 total time=  27.8s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=12;, score=0.887 total time=  29.4s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=12;, score=0.870 total time=  29.1s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=12;, score=0.909 total time=  29.5s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=14;, score=0.880 total time=  28.6s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=14;, score=0.878 total time=  29.3s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=14;, score=0.912 total time=  29.4s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=16;, score=0.875 total time=  30.5s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=16;, score=0.883 total time=  30.6s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=16;, score=0.906 total time=  30.6s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=18;, score=0.879 total time=  31.0s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=18;, score=0.919 total time=  30.7s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=18;, score=0.867 total time=  33.2s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.864 total time=  31.5s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.851 total time=  29.9s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.892 total time=  29.4s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.864 total time=  30.6s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.850 total time=  29.9s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.893 total time=  29.6s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.862 total time=  30.2s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.853 total time=  31.1s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=16;, score=0.866 total time=  30.8s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.892 total time=  31.0s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=16;, score=0.893 total time=  29.5s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=16;, score=0.850 total time=  32.0s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=18;, score=0.850 total time=  30.4s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=18;, score=0.862 total time=  31.2s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=18;, score=0.894 total time=  31.0s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.881 total time=  30.6s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.870 total time=  30.0s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.921 total time=  30.5s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=14;, score=0.884 total time=  28.0s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.870 total time=  30.9s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.875 total time=  31.2s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.916 total time=  29.6s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=16;, score=0.875 total time=  28.0s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=14;, score=0.875 total time=  30.0s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=16;, score=0.881 total time=  29.8s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=14;, score=0.912 total time=  30.3s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=16;, score=0.915 total time=  27.7s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=18;, score=0.879 total time=  27.7s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=18;, score=0.872 total time=  28.6s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=18;, score=0.912 total time=  29.4s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=10;, score=0.881 total time=  29.6s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=10;, score=0.916 total time=  29.0s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=12;, score=0.878 total time=  28.8s
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[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=10;, score=0.871 total time=  30.0s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=14;, score=0.882 total time=  28.4s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=12;, score=0.865 total time=  31.1s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=12;, score=0.916 total time=  30.9s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=14;, score=0.875 total time=  29.7s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=16;, score=0.880 total time=  29.5s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=16;, score=0.904 total time=  29.5s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=14;, score=0.914 total time=  31.8s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=16;, score=0.878 total time=  32.3s
[CV 3/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=18;, score=0.916 total time=  29.6s
[CV 2/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=18;, score=0.878 total time=  31.6s
[CV 1/3] END regressor__depth=4, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=18;, score=0.886 total time=  33.6s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=10;, score=0.884 total time=  34.2s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=10;, score=0.909 total time=  33.9s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=10;, score=0.879 total time=  34.6s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=12;, score=0.880 total time=  32.4s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=12;, score=0.884 total time=  35.0s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=12;, score=0.912 total time=  34.6s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=14;, score=0.879 total time=  32.6s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=14;, score=0.881 total time=  35.0s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=14;, score=0.912 total time=  34.1s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=16;, score=0.881 total time=  34.8s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=16;, score=0.884 total time=  35.8s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=16;, score=0.913 total time=  34.7s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=18;, score=0.912 total time=  31.8s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=18;, score=0.883 total time=  35.5s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=18;, score=0.881 total time=  35.1s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=10;, score=0.888 total time=  33.2s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=10;, score=0.891 total time=  33.3s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=12;, score=0.886 total time=  33.5s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=12;, score=0.919 total time=  33.4s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=10;, score=0.913 total time=  35.1s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=12;, score=0.893 total time=  35.5s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=14;, score=0.880 total time=  34.0s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=14;, score=0.911 total time=  33.3s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=14;, score=0.887 total time=  34.6s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=16;, score=0.894 total time=  33.6s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=16;, score=0.921 total time=  32.5s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=16;, score=0.890 total time=  35.2s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=10;, score=0.898 total time=  32.5s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=18;, score=0.917 total time=  33.3s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=18;, score=0.880 total time=  34.1s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=18;, score=0.887 total time=  35.4s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=10;, score=0.877 total time=  32.6s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=10;, score=0.916 total time=  32.9s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=12;, score=0.892 total time=  32.1s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=12;, score=0.916 total time=  31.5s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=12;, score=0.874 total time=  32.2s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=14;, score=0.896 total time=  32.4s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=14;, score=0.906 total time=  34.2s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=14;, score=0.887 total time=  35.6s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=16;, score=0.887 total time=  35.2s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=16;, score=0.889 total time=  33.6s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=16;, score=0.919 total time=  35.1s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=18;, score=0.902 total time=  34.0s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=18;, score=0.915 total time=  34.1s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=1, regressor__learning_rate=0.1, selector__num_features=18;, score=0.883 total time=  35.8s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=10;, score=0.882 total time=  34.0s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=12;, score=0.876 total time=  31.7s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=10;, score=0.877 total time=  34.3s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=10;, score=0.906 total time=  34.7s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=14;, score=0.878 total time=  32.7s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=12;, score=0.876 total time=  35.7s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=12;, score=0.907 total time=  35.3s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=14;, score=0.876 total time=  33.2s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=14;, score=0.907 total time=  32.9s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=16;, score=0.876 total time=  33.0s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=16;, score=0.880 total time=  34.3s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=16;, score=0.911 total time=  33.5s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=10;, score=0.895 total time=  29.5s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=18;, score=0.880 total time=  32.9s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=10;, score=0.887 total time=  31.6s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=12;, score=0.887 total time=  30.9s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=12;, score=0.883 total time=  30.4s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=18;, score=0.878 total time=  35.4s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=18;, score=0.907 total time=  34.7s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=10;, score=0.918 total time=  33.4s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=12;, score=0.918 total time=  28.9s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=14;, score=0.885 total time=  28.5s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=16;, score=0.890 total time=  30.6s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=14;, score=0.885 total time=  32.9s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=14;, score=0.916 total time=  31.5s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=16;, score=0.922 total time=  30.9s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=18;, score=0.888 total time=  30.7s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=16;, score=0.885 total time=  33.7s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=18;, score=0.885 total time=  29.1s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=10;, score=0.917 total time=  27.0s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=18;, score=0.922 total time=  31.6s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=10;, score=0.890 total time=  30.5s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=10;, score=0.893 total time=  30.8s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=12;, score=0.896 total time=  30.2s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=12;, score=0.915 total time=  28.3s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=12;, score=0.873 total time=  30.7s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=14;, score=0.888 total time=  26.8s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=14;, score=0.884 total time=  25.4s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=14;, score=0.912 total time=  26.5s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=16;, score=0.885 total time=  25.3s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=18;, score=0.893 total time=  26.3s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=16;, score=0.881 total time=  27.5s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=18;, score=0.890 total time=  26.2s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=16;, score=0.915 total time=  27.4s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=18;, score=0.917 total time=  23.7s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.876 total time=  24.7s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.902 total time=  24.6s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.872 total time=  26.5s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.903 total time=  26.3s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.873 total time=  27.2s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.872 total time=  28.8s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.874 total time=  29.2s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.869 total time=  27.0s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.903 total time=  26.4s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=16;, score=0.877 total time=  27.9s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=16;, score=0.871 total time=  27.8s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=18;, score=0.877 total time=  27.4s
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[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=18;, score=0.901 total time=  28.8s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.889 total time=  28.3s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.886 total time=  27.7s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.915 total time=  26.0s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.884 total time=  28.8s
[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.891 total time=  25.4s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.918 total time=  27.2s
[CV 1/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=14;, score=0.883 total time=  29.7s
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[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=16;, score=0.886 total time=  27.1s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=16;, score=0.923 total time=  25.2s
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[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=18;, score=0.888 total time=  26.8s
[CV 3/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=18;, score=0.921 total time=  26.3s
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[CV 2/3] END regressor__depth=6, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=10;, score=0.876 total time=  29.0s
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[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=1, regressor__learning_rate=0.01, selector__num_features=14;, score=0.915 total time=  35.3s
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[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=1, regressor__learning_rate=0.05, selector__num_features=14;, score=0.880 total time=  35.9s
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[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=12;, score=0.879 total time=  30.5s
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[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=3, regressor__learning_rate=0.01, selector__num_features=14;, score=0.880 total time=  35.1s
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[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=3, regressor__learning_rate=0.05, selector__num_features=16;, score=0.891 total time=  30.3s
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[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=16;, score=0.914 total time=  38.1s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=18;, score=0.884 total time=  38.8s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=3, regressor__learning_rate=0.1, selector__num_features=18;, score=0.924 total time=  39.0s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.872 total time=  37.5s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.907 total time=  35.4s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=10;, score=0.881 total time=  41.3s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.878 total time=  37.8s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.872 total time=  43.6s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.877 total time=  41.1s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=12;, score=0.904 total time=  44.2s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.872 total time=  42.6s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=16;, score=0.879 total time=  42.6s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=14;, score=0.904 total time=  46.1s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=16;, score=0.875 total time=  44.5s
```
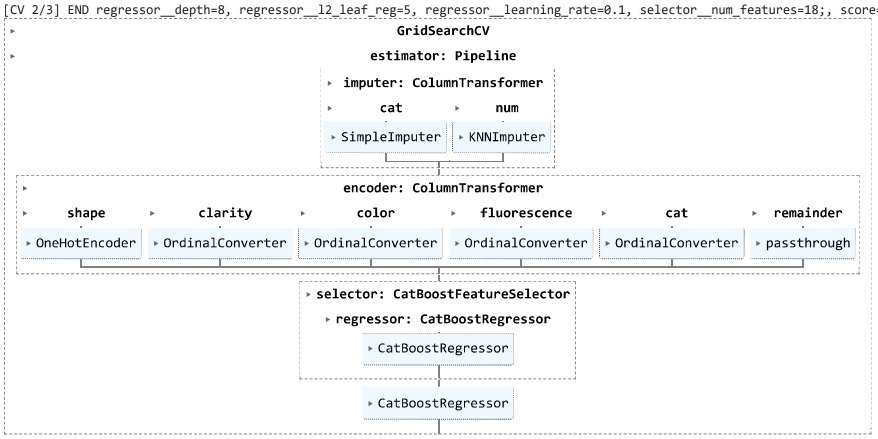

```
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=16;, score=0.905 total time=  45.1s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=18;, score=0.873 total time=  43.9s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=18;, score=0.908 total time=  44.1s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.01, selector__num_features=18;, score=0.875 total time=  45.8s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.894 total time=  45.5s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.885 total time=  46.0s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=10;, score=0.915 total time=  45.3s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.889 total time=  44.9s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.889 total time=  46.5s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=12;, score=0.913 total time=  44.4s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=14;, score=0.895 total time=  46.1s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=14;, score=0.894 total time=  46.7s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=14;, score=0.920 total time=  45.1s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=16;, score=0.892 total time=  38.1s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=16;, score=0.892 total time=  42.8s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=18;, score=0.887 total time=  42.8s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=18;, score=0.890 total time=  39.4s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=18;, score=0.921 total time=  38.0s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=10;, score=0.886 total time=  37.2s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.05, selector__num_features=16;, score=0.915 total time=  48.4s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=10;, score=0.882 total time=  35.6s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=10;, score=0.914 total time=  33.4s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=12;, score=0.893 total time=  36.6s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=12;, score=0.891 total time=  41.0s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=14;, score=0.890 total time=  37.2s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=14;, score=0.885 total time=  39.0s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=14;, score=0.911 total time=  37.9s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=12;, score=0.919 total time=  42.9s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=16;, score=0.895 total time=  39.1s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=16;, score=0.881 total time=  31.7s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=16;, score=0.914 total time=  28.0s
[CV 1/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=18;, score=0.892 total time=  27.0s
[CV 3/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=18;, score=0.922 total time=  28.3s
[CV 2/3] END regressor__depth=8, regressor__l2_leaf_reg=5, regressor__learning_rate=0.1, selector__num_features=18;, score=0.888 total time=  30.5s
```

Out[ ]:



In [ ]:
```
# Print the best hyperparameters and corresponding score
print(f"Best score: {grid_search.best_score_:.4f}")
print(f"Best parameters: {grid_search.best_params_}")
```

```
Best score: 0.9029
Best parameters: {'regressor__depth': 8, 'regressor__l2_leaf_reg': 5, 'regressor__learning_rate': 0.05, 'selector__num_features': 14}
```

By applying this, we improve the score of our model and above parameters are best tuned for our model

Final pipeline 🚀

We will use best parameters and build our final pipeline model

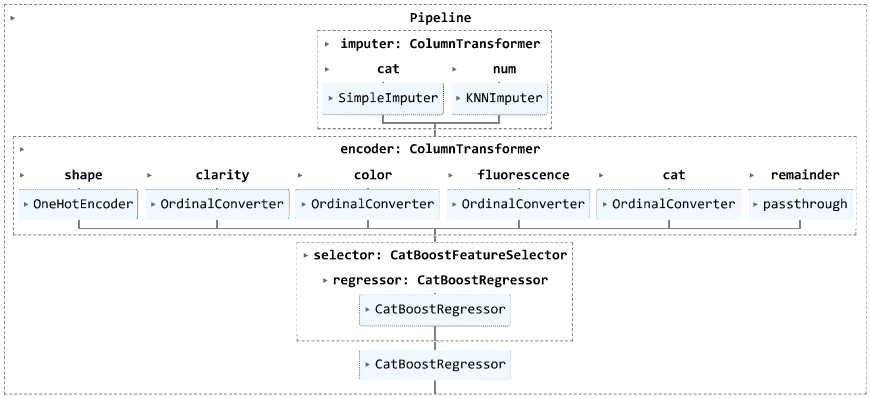
In [ ]:
```
imputer = ColumnTransformer(
    transformers=[
        (
            "cat",
            SimpleImputer(strategy="most_frequent"),
            ["Shape", "Clarity", "Colour", "Fluorescence", "Cut", "Polish", "Symmetry"],
        ),
        ("num", KNNImputer(n_neighbors=5), ["Weight", "Length", "Width", "Depth"]),
    ]
)

encoder = ColumnTransformer(
    transformers=[
        ("shape", OneHotEncoder(drop="if_binary", dtype=int), [0]),
        (
            "clarity",
            OrdinalConverter(
                ["I3", "I2", "I1", "SI2", "SI1", "VS2", "VS1", "VVS2", "VVS1", "IF"]
            ),
            [1],
        ),
        (
            "color",
            OrdinalConverter(
                [
                    "Y-Z",
                    "W-X",
                    "W",
                    "U-V",
                    "S-T",
                    "Q-R",
                    "O-P",
                    "O",
                    "N",
                    "M",
                    "L",
                    "K",
                    "J",
                    "I",
                    "H",
                    "G",
                    "F",
                    "E",
                    "D",
                    "FANCY",
                ]
            ),
            [2],
        ),
```

```
        (
            "fluorescence",
            OrdinalConverter(["N", "M", "F", "SL", "ST", "VS", "VSL"]),
            [3],
        ),
        ("cat", OrdinalConverter(["FR", "GD", "VG", "EX"]), [4, 5, 6]),
    ],
    remainder="passthrough",
)

selector = CatBoostFeatureSelector(
    regressor=CatBoostRegressor(verbose=0, allow_writing_files=False), num_features=14
)

regressor = CatBoostRegressor(
    verbose=0, allow_writing_files=False, depth=8, l2_leaf_reg=5, learning_rate=0.05
)


pipeline = Pipeline(
    steps=[
        ("imputer", imputer),
        ("encoder", encoder),
        ("selector", selector),
        ("regressor", regressor),
    ]
)
```

Train our model

In [ ]:
```
pipeline.fit(X_train, y_train)
```

Out[ ]:



Testing the model using train dataset

In [ ]:
```
pipeline.score(X_train, y_train)
```

Out[ ]:
```
0.9916820007381009
```

Testing the model using test dataset

In [ ]:
```
pipeline.score(X_test, y_test)
```

Out[ ]:
```
0.8802009811349544
```

Saving the model

Save the model using dill

In [ ]:
```
# Save the trained pipeline to a file
with open("../data/model.br", "wb") as f:
    dill.dump(pipeline, f)
```

Testing our model

We test the model be the user

Loading model

Next, we will load the model using dill.

In [ ]:
```
with open("../data/model.br", "rb") as f:
    model = dill.load(f)
```

Create a static user input

In [ ]:
```
# Create a pandas dataframe from the input data
input_data = [
    "PRINCESS",
    0.3100586,
    "VVS1",
    "D",
    "EX",
    "EX",
    "GD",
    "N",
    3.8496094,
    3.7695312,
    2.6894531,
]
columns = [
    "Shape",
    "Weight",
    "Clarity",
    "Colour",
    "Cut",
    "Polish",
    "Symmetry",
    "Fluorescence",
    "Length",
```

```
    "Width",
    "Depth",
]

input_df = pd.DataFrame([input_data], columns=columns)
```

Predict the output

In [ ]:
```
model.predict(input_df)
```

Out[ ]:
```
array([1085.27686875])
```

Create user interface

In [ ]:
```
# Define the input form
print("Please enter the following details for the diamond:")
shape = str(
    input(
        "Shape ('PRINCESS', 'ROUND', 'HEART', 'OVAL', 'CUSHION', 'EMERALD', 'MARQUISE', 'PEAR'): "
    )
)
weight = float(input("Carat weight (e.g. 1.23): "))
clarity = str(
    input(
        "Clarity ('I3', 'I2', 'I1', 'SI2', 'SI1', 'VS2', 'VS1', 'VVS2', 'VVS1', 'IF'): "
    )
)
color = str(
    input(
        "Color ('Y-Z', 'W-X', 'W', 'U-V', 'S-T', 'Q-R', 'O-P', 'O', 'N', 'M', 'L', 'K', 'J', 'I', 'H', 'G', 'F', 'E', 'D', 'FANCY'): "
    )
)
cut = str(input("Cut ('FR', 'GD', 'VG', 'EX'): "))
polish = str(input("Polish ('FR', 'GD', 'VG', 'EX'): "))
symmetry = str(input("Symmetry ('FR', 'GD', 'VG', 'EX'): "))
fluorescence = str(input("Fluorescence ('N', 'M', 'F', 'SL', 'ST', 'VS', 'VSL'): "))
x = float(input("Length in mm (e.g. 6.55): "))
y = float(input("Width in mm (e.g. 6.51): "))
z = float(input("Depth in mm (e.g. 4.11): "))
```

```
Please enter the following details for the diamond:
```

In [ ]:
```
input_data = [
    shape,
    weight,
    clarity,
    color,
    cut,
    polish,
    symmetry,
    fluorescence,
    x,
    y,
    z,
]
columns = [
    "Shape",
    "Weight",
    "Clarity",
    "Colour",
    "Cut",
    "Polish",
    "Symmetry",
    "Fluorescence",
    "Length",
    "Width",
    "Depth",
]
```

In [ ]:
```
# Create a pandas dataframe from the input data
input_df = pd.DataFrame([input_data], columns=columns)
```

In [ ]:
```
output = model.predict(input_df)
```

In [ ]:
```
# Print the predicted output
print("The predicted price of the diamond is $", round(output[0], 2))
```

```
The predicted price of the diamond is $ 11272.6
```

Conclusion 🎉

In this notebook, we build our pipeline model that consists of several transformers and an estimator. It is designed to preprocess and model diamond price data. The pipeline has four main stages, which are as follows:

- `imputer` : This transformer replaces missing values in the dataset. It contains two imputers - one for categorical features (using the most frequent value) and another for numeric features (using the KNN algorithm).
- `encoder` : This transformer encodes the categorical features using OneHotEncoder for the Shape feature and an OrdinalConverter for the remaining categorical features.
- `selector` : This transformer is a custom transformer that uses a CatBoostRegressor model to identify the most important features in the dataset. The transformer returns the top 14 features based on their importance.
- `regressor` : This is the final stage of the pipeline and it contains the CatBoostRegressor model. The model is used to predict the diamond prices using the selected features.

We also applied hyper tuning using GridSearchCV for our model to improve its performances and saved our model in pickle format

Finally, we also implemented user interface to test and demonstrate the model.